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Electrochromism in indium-tin-oxide films for laser-writing application

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ABSTRACT

We report for the first time laser-writing application on an electrochemically NH_4^+ ion intercalated indium-tin oxide (ITO). NH_4^+ ion intercalation results into a color change from transparent to brown with a significant enhancement of the surface roughness. Both bare and intercalated ITO's are explored for their structural, surface morphological, and wettability studies. Upon NH_4^+ intercalation, both an optical absorbance and the surface wettability of ITO changes substantially. A micrometer-scale laser-writing application on intercalated ITO film is achieved after exposing an Ar-ion laser that is non-erasable even after a period of six months. This idea can be used for maintaining long-term safe secrete data.

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1. Introduction

Indium-tin-oxide (ITO), a wide band-gap semiconductor ($E > 3.5$ eV), is deposited as a thin film on a glass substrate and used as an auxiliary anode in switchable mirrors and smart windows [1]. The intercalation of ITO with lithium [2,3], sodium [4], and hydrogen [5,6] has been well studied in the literature because of its favorable inter-atomic spacing. ITO is optically transparent electrode in the visible region and has variety of applications in liquid-crystal displays, solar cells, energy-efficient windows, spectroelectrochemistry, and organic light-emitting diodes [7,8]. Conductivity of In_2O_3 with an appropriate Sn doping is metallic. This is because; tin has four valence electrons, and is therefore an electron donor in indium oxide when become a part of the In lattice. The electrical properties of ITO depend on the tin concentration, which induces n-type doping, so that ITO in general exhibits electrical conductivity, and hence the conductivity of ITO is found to increase with the tin concentration, but a very high concentration of tin normally results in a drastic reduction in conductivity [9]. Nevertheless, ITO is a non-stoichiometric oxide, and the contribution of oxygen vacancies to its conductivity is noticeable. It is widely assumed that the Sn component of ITO is present as Sn(IV) and Sn(II) oxides. In sputter-deposited ITO films, even when using an oxide target, an annealing step in an oxidizing atmosphere is required to achieve high conductivity. This is because the oxygen vacancies are filled and are redistributed throughout the ITO lattice, and this changes the ratio of Sn(IV)/Sn(II) [10].

In continuation to our research on ITO substrate [11], in the present investigation we explore different applications other than normal routine wherein NH_4^+ ions are intercalated into an ITO polished, purchased from the Samsung Corning Corporation (Korea) with the following specifications: flatness = 227 Å, transmission = 87.9%, sheet resistance = 9.2 Ω/cm^2, and film thickness = 185 nm and commercial ammonium hydroxide without further purification, NH_4OH (28%) (Junsei Chemical Co. Ltd., Japan), to intercalate NH_4^+ ions into the ITO films by using a galvanostatic method. The pH was adjusted to 10 by adding acidic Milli-Q water to the ammonium hydroxide solution. A three-electrode one-compartment 100 ml electrochemical cell with a 1 cm^2 Pt counter electrode and an Ag/AgCl reference electrode was used in our experiments. The bath temperature was maintained at 303 K. A cathodic current of 2 mA/cm^2 was passed for a period of 300 s. After about 10 s, the ITO film started to show change in color, which was complete after a period of 300 s. A further increase in intercalation time resulted in ITO leaching. In short, we observed that using the NH_4OH solution in the presence of an oxidizing agent turned the transparent ITO layer to brown if it was negatively biased.

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2. Experimental details

Electrochemical intercalation leads to electrochromism, and it has been reported that both cathodic and anodic electrochromism lead to a change of color in transition metal oxides [12]. Permanent quasi-topotactic reactions have not been investigated in detail. Ammonium hydroxide solution contains NH_4^+ and OH^- ions. The ionic radius of the NH_4^+ ion (1.48 Å) [13] is considerably smaller than the interplanar *d* spacing of ITO (2.19–2.60 Å) [14], and the intercalation of NH_4^+ seems to be favorable. Contact-mode atomic force microscopy (AFM) was used to obtain two- and three-dimensional (2D and 3D, respectively) images. The scan rate used was 0.95 Hz, and the contact area was $1 \times 1\,\mu m^2$. AFM analysis was carried out on ITO films with and without intercalation of NH^{4+} ions. The structural analysis was carried out with Philips X-ray diffractometer (PW-3710) using copper target of wavelength 1.5405 Å. To verify the color change, the UV–vis spectra of ITO with and without intercalation were recorded using a Model 100 Conc spectrophotometer (Cary, 100 Victoria, Australia). For surface-induced superhydrophilicity study, a water contact angle was measured using contact angle analyzer (SEO–PHOENIX 150, Korea). This is based on measuring the contact angle of test liquid (de-ionized water) on the solid surface, in present case ITO and NH_4^+ ion intercalated ITO. For supporting the surface modification of ITO after NH^{4+} intercalation, room temperature (25 °C) cyclic-voltammetry measurement was carried out in a classical three-electrode cell with ITO and NH^{4+} intercalated ITO as working electrodes. The counter electrode was a Pt foil and the reference electrode was Ag/AgCl electrode, respectively. The electrolyte was tributyl ammonium salt solution.

3. Results and discussion

After intercalation, the 2D compact vertical uneven rod-like morphology changed to that of broad, well-defined elongated rods with narrow ends, without disturbing the original architecture as shown in Fig. 1(a) and (b), suggesting that ITO grains are swollen after NH_4^+ ion intercalation. Fig. 1(c) shows a 3D image of ITO composed of vertically grown connected plates within an area of $1 \times 1\,\mu m^2$, in addition to some hills and valleys with an average roughness of 25.12 Å, as measured from the wide-range histogram. However, after intercalation, the 3D surface morphology (Fig. 1(d)) is changed to that of relatively larger, separate rods with an increase in average roughness up to 105.75 Å, consistent with the conclusion made based on 2D images.

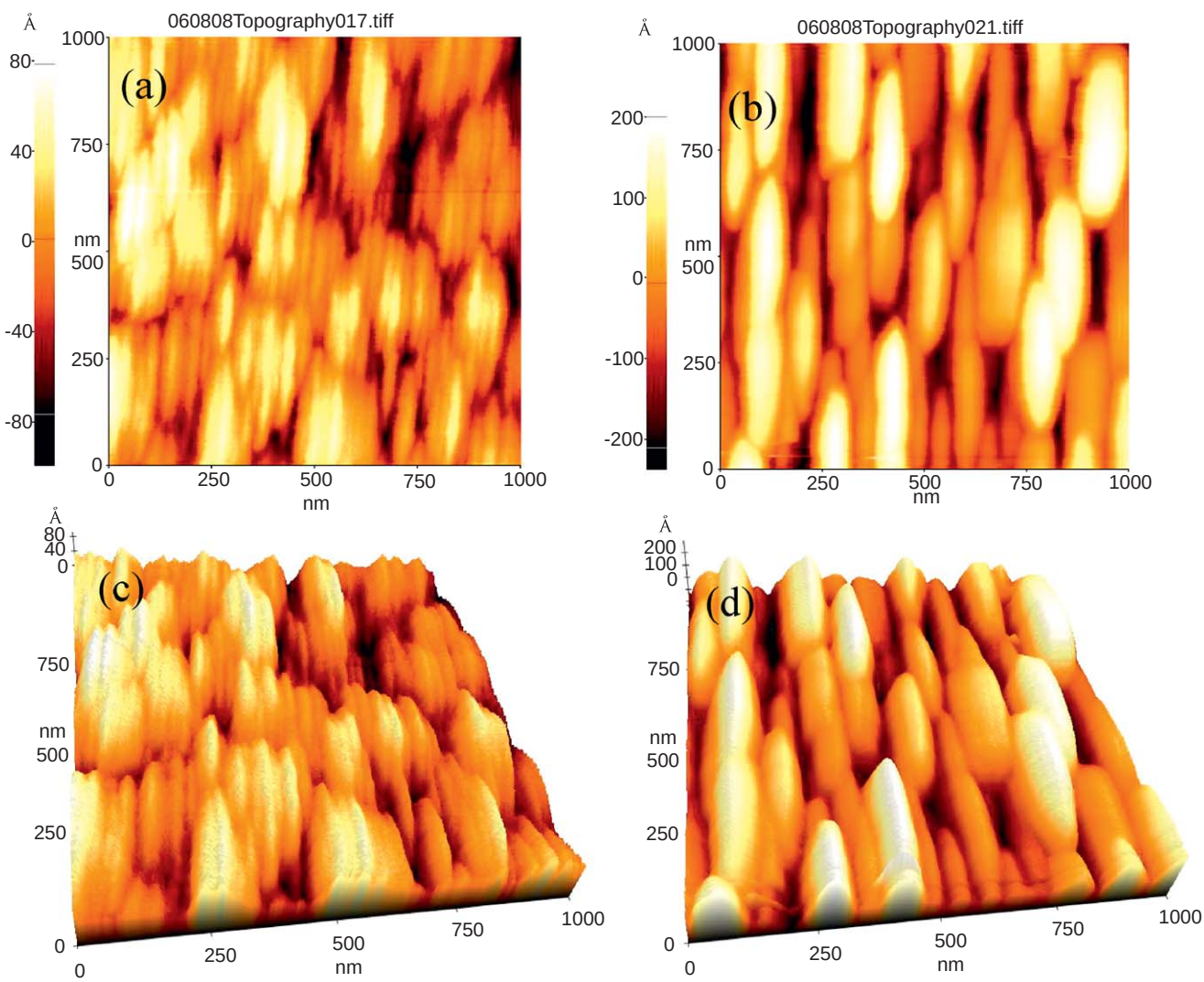

Fig. 1. Two-dimensional ((a) and (b)) and three-dimensional ((c) and (d)) contact-mode AFM images of bare and intercalated ITO, respectively. The 3D images show that the intercalation of ITO did not change the surface architecture, except to induce an average roughness in the range 25.12–105.75 Å.

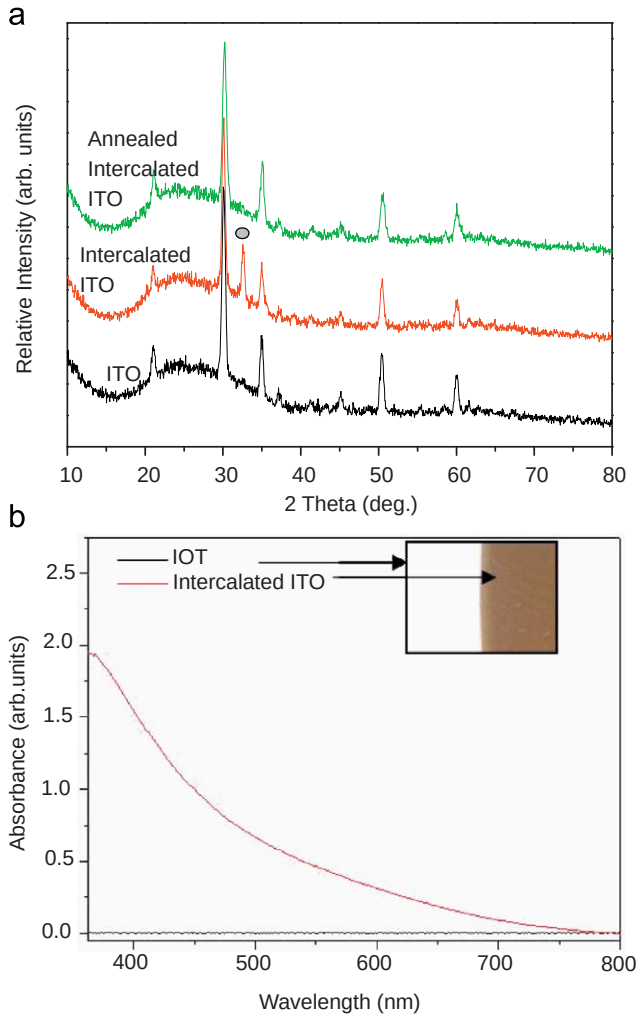

Fig. 2. (a) The XRD patterns of ITO, intercalated ITO and annealed intercalated ITO complex confirming the presence of ammonium only in intercalated ITO as marked by filled circle and (b) the absorption spectra of bare and intercalated ITO samples, where a shift towards the visible region is shown. The inset shows the brown coloration of ITO after NH_4^+ intercalation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

X-ray spectra corresponding to ITO, intercalated ITO and annealed intercalated ITO (250 °C for 1 h) are presented in Fig. 2(a). The sharp peak marked by filled circle at 32.55° corresponds to ammonium azide (JCPDF data file: 09-0409). This interesting observation is noticed only for the intercalated ITO, confirming intercalation of ammonium ions into ITO matrix. Further, after annealing, it is noteworthy to note that peak (1 0 3) get disappeared due to the local heating as the melting temperature of ammonium azide is 160 °C. From Fig. 2(b), it is shown that the ITO film does not have any absorption bands in the visible region. However, in case of intercalated ITO films, the fundamental absorption began at a wavelength of 780 nm. We believed this must be a first report where significant enhancement in the visible light absorption due to the intercalation of NH_4^+ ions, compared with non-intercalated ITO, is obtained. The presence of NH_4^+ ions is a prerequisite for inducing a brown coloration in the ITO (inset in Fig. 2), that cannot be attributed to light scattering. The cyclic-voltammetry response of ITO is changed after NH^{4+} ions intercalation, indicating that there is a shift in conduction band closer to vacuum which is also confirmed from enhanced current density and the appearance of new peak at −0.72 V (Fig. 3). This behavior reveals that the change in surface

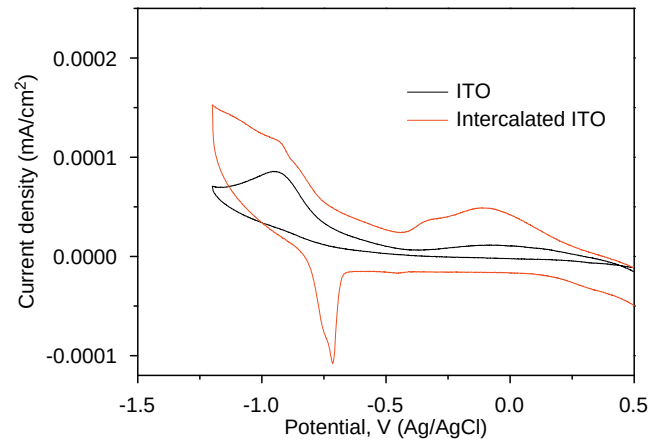

Fig. 3. Cyclic-voltammograms of bare and intercalated ITO are supporting the surface modification due to NH_4^+ ion intercalation.

architecture of ITO is due to the NH^{4+} ion intercalation. This remarkable and interesting feature of intercalated ITO would find different applications such as photoelectrochemical studies [15], water splitting [16,17], surface-induced superhydrophilicity [18], and extremely thin absorber solar cells [11,19]. The measured contact angle value is decreased from 48.45° to 9.60° (Fig. 4) after NH_4^+ ion intercalation due to the increase in surface roughness. The roughness itself is an essential characteristic of the surfaces. Water adhesion to rough surface may be effective because of the intrinsically high surface energy of atoms on the asperity surface. The increase in surface area, possibly by a very high factor, also raises the surface energy when expressed per unit nominal area [19]. The effect of internal and external stain energies on surface wettability is reported elsewhere [20].

A new, straightforward, stable, and low-cost innovative application of this intercalated ITO in laser writing was carried out using an Ar-ion laser operating at $\lambda = 775$ nm. In our experiments, a parabolic mirror was used to focus the laser light through a 50 × objective to a spot diameter of 1.13 μm. An incident power of 50 mW/cm^2 was measured by positioning a photodiode just in front of the intercalated ITO sample. Due care was taken while carrying manual laser writing. It is interesting to note that after a period of about 5 s, the ITO writing manifested itself as a dark writing on a white background. Local heating created by the absorption of the laser beam is evidenced from XRD pattern of annealed intercalated ITO that resulted in the decomposition of thin film $(NH_4N_3)_x$ $(ITO)_{1-x}$ complex, changing the local color of the sample to black [21]. As a test, we wrote the word, "HANYANG" the name of our research institute in micrometer scale on an intercalated ITO complex, which is non-erasable, even after six months of time span (Fig. 5).

4. Conclusions

In nutshell, here, for the first time, we demonstrated laser-writing application on NH_4^+ ion intercalated ITO. Due to this phenomenon, there is a color change of ITO from transparent to brown with a significant enhancement of the surface roughness and the optical absorption. Increase in roughness, 105.75 Å in an intercalated ITO complex exerts a significant influence on the water adhesion by producing superhydrophilic (∼10°) and UV–vis absorbing surface, making it possible to be used in visible light photocatalysis, similar to $CdIn_2S_4$ [22]. A micrometer-scale laser writing on intercalated ITO complex is found to be non-erasable even after long-term safe depositing confidential data.

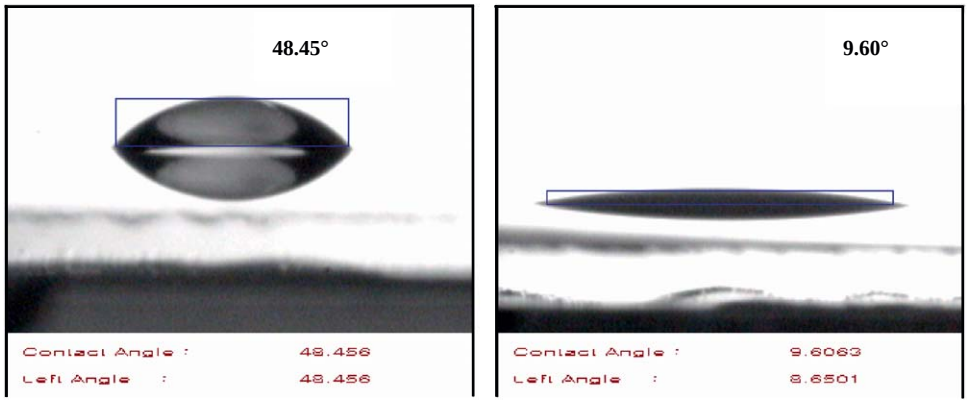

Fig. 4. Water contact angle measurements on ITO and intercalated ITO surfaces showing superhydrophilic nature for intercalated ITO with ~10°.

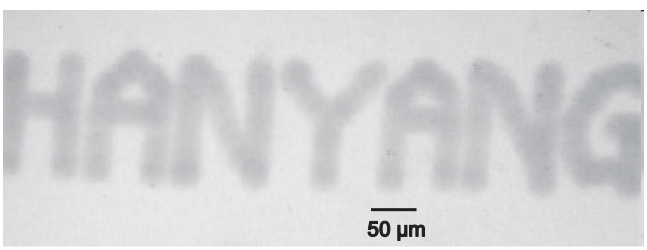

Fig. 5. An optical image of letters "HANYANG", the name of our institute, written on intercalated ITO used for laser writing in an area of approximately 572 μm^2.

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